

A is bus incidence matrix

e \ buses	(buses)				e \ bus
	1	2	3	4	
1	1	0	0	0	Tree branches
2	0	1	0	0	
3	0	0	1	0	
4	0	0	0	1	
5	1	0	0	-1	Link branches
6	0	0	-1	1	
7	0	1	-1	0	
8	1	0	-1	0	
9	1	-1	0	0	

$e = \text{no. of elements}$
 $n = \text{no. of buses}$

$a_{ip} = 1$ if i th element is incident to and oriented away from p th bus.

$a_{ip} = -1$ if i th element is incident to and oriented towards p th bus

$a_{ip} = 0$ if i th element is not incident to p th bus

A_b is of order $(b \times n)$

A_L is " $(l \times n)$

b is the no. of tree branches

l is the no. of link branches

Formulation of Y_{bus} and Z_{bus}

$$I + J = YV$$

$$\text{and } V = AV_{bus}$$

$$I + J = YAV_{bus}$$

pre multiplying A^t

$$A^t I + A^t J = A^t Y A V_{bus}$$

Since A shows the incidence of elements to buses, $A^t I$ is a scalar sum in which each element is the algebraic sum of currents through the network elements terminating at a bus. As per Kirchhoff's law, the algebraic sum of currents at a sum is zero.

$$A^t I = 0$$

Similarly $A^t J$ gives the algebraic sum of source currents at each bus.

$$A^t J = I_{bus}$$

$$I_{bus} = A^t Y A V_{bus}$$

$$I_{bus} = [A]^t Y_{bus} V_{bus}$$

$$[A]^t \Rightarrow Y_{bus} = A^t Y A$$

and $Z_{bus} = Y_{bus}^{-1}$

$$Z_{bus} = [Y_{bus}]^{-1}$$

Load Flow Equations

Recalling that,

$I_{bus} = [Y_{bus}]V_{bus}$, the current entering the i th bus of n bus system is given by

$$I_i = Y_{i1}V_1 + Y_{i2}V_2 + \dots + Y_{ii}V_i + \dots + Y_{in}V_n$$

$$= \sum_{p=1}^n Y_{ip}V_p$$

$$= \sum_{p=1}^n |Y_{ip}| |V_p| \angle (\delta_p + \gamma_{ip})$$

Where $V_p = |V_p| \angle \delta_p$ and $Y_{ip} = |Y_{ip}| \angle \gamma_{ip}$

The complex power injected into i th bus is

$$S_i = P_i + jQ_i = V_i I_i^*$$

The load flow problem is handled more easily if I_i (rather than I_i^*) is used.

Taking the complex conjugate

$$S_i^* = P_i - jQ_i = V_i^* I_i, \quad i = 1, 2, \dots, n$$

$$\text{or } S_i^* = P_i - jQ_i = V_i^* \sum_{p=1}^n Y_{ip} V_p$$

$$\text{or } P_i - jQ_i = |V_i| \sum_{p=1}^n |Y_{ip}| |V_p| \angle (\delta_p + \gamma_{ip} - \delta_i)$$

$$P_i - jQ_i = |V_i| \sum_{p=1}^n |Y_{ip}| |V_p| \angle (\delta_i - \delta_p - \gamma_{ip})$$

$$V_i = |V_i| \angle \delta_i, \quad V_i^* = |V_i| \angle -\delta_i, \quad Y_{ip} = |Y_{ip}| \angle \gamma_{ip}$$

Separating real and imaginary parts.

$$P_i = |V_i| \sum_{p=1}^n |Y_{ip}| |V_p| \cos (\delta_i - \gamma_{ip} - \delta_p)$$

$$Q_i = |V_i| \sum_{p=1}^n |Y_{ip}| |V_p| \sin (\delta_i - \gamma_{ip} - \delta_p)$$

These equations are known as steady state load flow equations. These equations are nonlinear. For each of the n system buses we have two such eqns giving a total of $2n$ eqns. At each bus we have 4 variables P_i , Q_i , V_i and δ_i giving a total no. of $4n$ variables.

bus type	specified quantities	unknown quantities
load or PQ bus	P_i, Q_i	V_i, δ_i 85%
generator bus or voltage control bus or PV bus	P_i, V_i	Q_i, δ_i 15
Ref. bus or slack bus or swing bus	V_i $\delta_i = 0$	P_i, Q_i 1
generally the bus of the largest generating station		

The solution of the load flow problem consists of assuming a certain initial bus load configuration, specifying the $2n$ known variables and using some numerical method to find the remaining $2n$ variables for the system with known Y_{bus} . The final solution must satisfy some constraints. These constraints are:

- (1) Voltage magnitude at various buses must be within limits
- (2) active and reactive generator power at different buses must be within the minimum and maximum limits
- (3) total generation must be equal to the total load plus losses.
- (4) The system stability consideration impose a limit on max. value of δ_i .

Gauss-Seidel Method for Load Flow Study

Advantages

- 1- Simplicity of technique
- 2- Small computer memory requirement.
- 3- Less computational time per iteration.

Disadvantage

- 1- Slow rate of convergence, large number of iterations
- 2- No. of iterations increase directly with the increase of no. of buses.
- 3- Effect on convergence due to choice of slack bus

G-S Method When P-V buses are Absent

Example

Obtain the value of δ and γ after 4 G-S iterations

$$\begin{aligned}\delta &= 0.7 \sin \delta + 0.2 \cos \gamma \\ \gamma &= 0.7 \cos \delta - 0.2 \sin \gamma\end{aligned}$$

$$\text{Assume } \delta^0 = \gamma^0 = 0.5$$

Sol

$$\delta^0 = 0.5$$

$$\gamma^0 = 0.5$$

$$\delta_1 = 0.7 \sin(0.5) + 0.2 \cos(0.5) = 0.511143893$$

$$\gamma_1 = 0.7 \sin(0.511143893) - 0.2 \cos(0.5) = 0.5146548547$$

$$\delta_2 = 0.7 \sin(0.511143893) + 0.2 \cos(0.5146548547) = 0.5164971916$$

$$\gamma_2 = 0.7 \sin(0.5164971916) - 0.2 \cos(0.5146548547) = 0.51024$$

$$\delta_3 = 0.5202114603$$

$$\gamma_3 = 0.5097223258$$

$$\delta_4 = 0.5225205507 \checkmark$$

$$\gamma_4 = 0.509007831 \checkmark$$

values after

G-S Method when P-V Buses are Absent

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This implies that out of 'n' buses, one bus is the slack bus and the remaining $n-1$ buses are P-Q buses. Initially, we assume the magnitude and angles at these $n-1$ buses, and update these voltages at every iteration.

$$I_i = \sum_{p=1}^n Y_{ip} V_p$$

$$= \sum_{\substack{p=1 \\ p \neq i}}^n Y_{ip} V_p + V_i Y_{ii}$$

$$\text{or } \cancel{V_i I_i} V_i Y_{ii} = I_i - \sum_{\substack{p=1 \\ p \neq i}}^n Y_{ip} V_p$$

$$\text{or } \cancel{V_i} = \frac{1}{Y_{ii}} \left[I_i - \sum_{\substack{p=1 \\ p \neq i}}^n Y_{ip} V_p \right]$$

$$\text{Also } I_i = \frac{P_i - jQ_i}{V_i^*}$$

$$\therefore V_i = \frac{1}{Y_{ii}} \left[\frac{P_i - jQ_i}{V_i^*} - \sum_{\substack{p=1 \\ p \neq i}}^n Y_{ip} V_p \right]$$

Since bus 1 is the slack bus, the above eqn. represents a set of $(n-1)$ eqns. for $i = 2, 3, \dots, n$ which are to be solved simultaneously for $V_2, V_3, \dots, V_n$. The voltages and admittances are complex quantities.

$$\text{or } V_i = \frac{K_i}{V_i^*} - \sum_{\substack{p=1 \\ p \neq i}}^n L_{ip} V_p \quad \text{--- (A)}$$

$$\text{where } K_i = \frac{P_i - jQ_i}{Y_{ii}} \quad \text{for } i = 2, 3, \dots, n$$

$$L_{ip} = \frac{Y_{ip}}{Y_{ii}} \quad \text{for } i = 2, 3, \dots, n \\ p = 1, 2, \dots, n \\ p \neq i$$

It is necessary to assume initial values of $V_2, V_3 \dots, V_n$. We know that the voltages at different buses in a power system are close to 1 p.u. Therefore we assume $V_2, V_3 \dots V_n$ to be 1+j0 initially. ~~The iterative process~~ At every step of iteration, we use the most ~~most~~ updated values of the bus voltages to compute the new values of the bus voltages.

When $(k+1)^{th}$ iteration is being performed to calculate V_i , the most updated values ^{of bus voltages are the values} obtained in $(k+1)^{th}$ iteration for all buses previous to i^{th} bus (i.e. for $p = 1, 2, \dots, i-1$) and the values obtained in k^{th} iteration for i^{th} bus and the buses after the i^{th} bus (i.e. $p = i, i+1, \dots, n$). In view of this, eqn. A4 can be written ~~for $(k+1)^{th}$~~ for $(k+1)^{th}$ iteration

$$V_i^{k+1} = \frac{K_i}{(V_i^k)^*} - \sum_{p=1}^{i-1} L_{ip} V_p^{k+1} - \sum_{p=i+1}^n L_{ip} V_p^k$$

$i = 2, 3, \dots, n$

the iterative process is continued till the changes in the bus voltages in successive iterations become less than a certain pre-specified tolerance for all buses

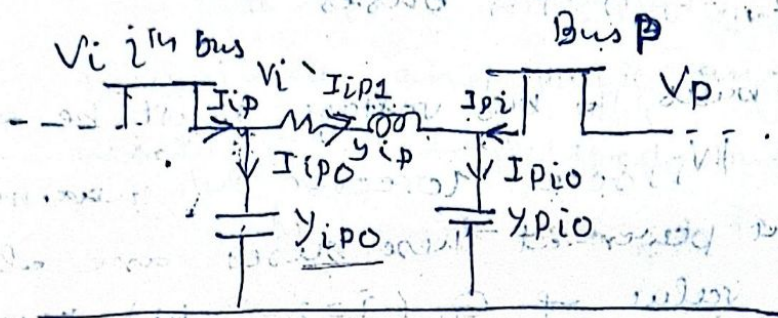
$$|\Delta V_i^{k+1}| = |V_i^{k+1} - V_i^k| < \epsilon$$

to reduce the no. of iteration acceleration factor α can be used $\alpha = 1.6$ for good value of load flow.

When all bus voltages have been determined S_i^* is obtained from eqn.

$$S_i^* = P_i - jQ_i = V_i^* \sum_{p=1}^n Y_{ip} V_p$$

The final step in load flow study is the calculation of line flows



The current $I_{cp} = I_{cp1} + I_{cp2} = (V_i - V_p) Y_{ip} + V_i Y_{ipo}$

~~$$S_{cp} = P_{cp} + jQ_{cp}$$~~

$$= V_i I_{cp}^* = V_i (V_i - V_p)^* Y_{ip}^* + V_i V_i^* Y_{ipo}^*$$

Similarly

$$S_{pi} = V_p (V_p - V_i)^* Y_{ip}^* + V_p V_p^* Y_{pio}^*$$

The ^{flow} power loss in the line connecting i th and p th buses is given by $S_{cp} + S_{pi}$

Modification of G-S Method when P-V buses are present

Some of the buses in a power system are P-V buses at which Q, δ are unknowns. So it is necessary to modify the above computational procedures.

Let the buses are

$i = 1$ slack bus

$i = 2, 3, \dots, m$ P-V bus

$i = m+1, m+2, \dots, n$ PQ buses.

For the voltage control buses, the bus voltage must be equal to specified voltages $|V_i|_{\text{spec}}$. Moreover the maximum and minimum reactive power at these buses are also specified and the value of Q_i ($i = 2 \dots m$) must be within these limits. Thus the conditions to be met

$$|V_i| = |V_i|_{\text{spec}} \quad i = 2 \dots m$$

$$Q_{\text{imin}} < Q_i < Q_{\text{imax}} \quad i = 2 \dots m$$

Q_i can be calculated from the following eqn.

$$Q_i = |V_i| \sum_{p=1}^n |V_p| |Y_{ip}| \sin(\delta_i - \gamma_{ip} - \delta_p)$$

At the start of $(k+1)^{\text{th}}$ iteration, $|V_i|^k$ is the magnitude of V_i obtained during k^{th} iteration may not necessarily be equal to $|V_i|_{\text{spec}}$ obtained during iteration we save this value of V_i^k .
 $|V_i|^k = |V_i|_{\text{spec}} / \delta_i^k$

The values of $|V_p|$ and δ_p to be used in eqn are those obtained during $(k+1)^{\text{th}}$ iterations (for $p = 1$ to $i-1$) and those obtained during k^{th} iteration (for $p = i$ to n). Moreover for every iteration $|V_i|$ must be equal to $|V_i|_{\text{spec}}$. In view of this eqn for Q_i will be

$$Q_i^{k+1} = |V_i|_{\text{spec}} \sum_{p=1}^{i-1} |Y_{ip}| |V_p|^{k+1} \sin(\delta_i^k - \gamma_{ip} - \delta_p^{k+1})$$

$$+ |V_i|_{\text{spec}} \sum_{p=i}^n |Y_{ip}| |V_p|^k \sin(\delta_i^k - \gamma_{ip} - \delta_p^k)$$

Now 1:- Check if $Q_i^{(k+1)}$ is within limits i.e. $Q_{\min} < Q_i^{k+1} < Q_{\max}$
 if so calculate new value of k_i and then
 compute $V_i^{(k+1)}$ using $|V_i|_{\text{spec}}$ and δ_i^k for the
 magnitude and phase angle δ_i^{k+1} of V_i^k .
 Reset $|V_i^{k+1}|$ to $|V_i|_{\text{spec}}$ but retain the phase
 angle δ_i^{k+1} and continue to the next bus

2- If $Q_i^{k+1} > Q_{\max}$ set $Q_i^{k+1} = Q_{\max}$ find the new
 value of k_i and treat the i th bus as P-Q bus
 Similarly if $Q_i^{k+1} < Q_{\min}$ set $Q_i^{k+1} = Q_{\min}$
 Find the new value of k_i and treat the
 i th bus as P-Q bus similarly for it

$$x_{\Delta} + x_{\Sigma} = x_c$$

$$\left[\frac{9.6}{10} \cos 7 \right] - x_{\Sigma} =$$

for it to stabilize (17.1.1)

$$\left[\frac{9.6}{10} \cos 7 \right] - x_{\Sigma} + x_{\Delta} + x_c =$$

the busbar voltage at

busbar voltage $V_b \geq V_{\min}$

Newton - Raphson Method

Consider the eqn.

$$f(x) = 0$$

Assume that x^0 is the initial estimate of x and Δx^0 is the correction. $(x^0 + \Delta x)$ will be the soln.

$$\therefore f(x^0 + \Delta x^0) = 0$$

Taylor series $\approx$

$$f(x^0 + \Delta x^0) = f(x^0) + \Delta x^0 \left(\frac{df}{dx} \right)^0 + \frac{1}{2} (\Delta x^0)^2 \left(\frac{d^2 f}{dx^2} \right)^0 + \dots$$

A first estimate of the correction is made by truncating the series after the first two terms.

$$f(x^0) + \Delta x^0 \left(\frac{df}{dx} \right)^0 \approx 0$$

$$\text{or } \Delta x^0 \approx - \frac{f(x^0)}{\left(\frac{df}{dx} \right)^0}$$

The first estimate of the root is

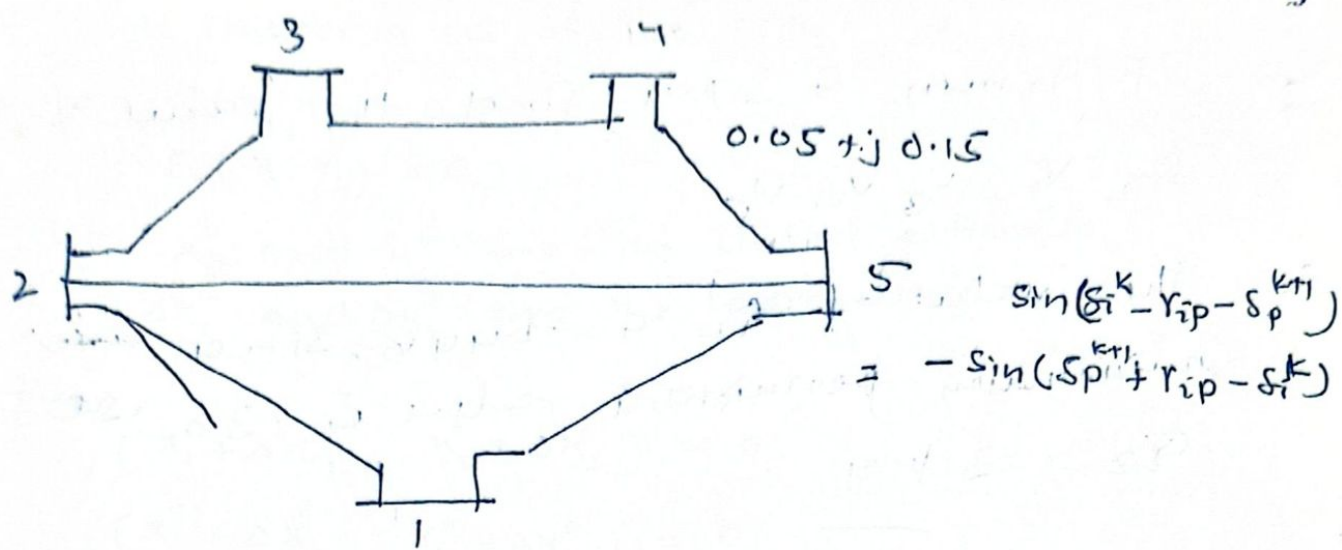
$$\begin{aligned} x^1 &= x^0 + \Delta x^0 \\ &= x^0 - \left[f(x^0) / \left(\frac{df}{dx} \right)^0 \right] \end{aligned}$$

The $(k+1)^{\text{th}}$ estimate of the root is

$$x^{k+1} = x^k + \Delta x^k = x^k - \left[f(x)^k / \left(\frac{df}{dx} \right)^k \right]$$

The iterations are continued till

$$|x^{k+1} - x^k| \leq \epsilon \quad (\text{specified tolerance})$$



$$\begin{aligned} V_2' &= -|V_2|_{spec} \left[|Y_{21}| |V_1| \sin(\delta_1^0 + \gamma_{21} - \delta_2^0) + |Y_{22}| |V_2| \sin(\delta_2^0 + \gamma_{22} - \delta_2^0) \right. \\ &\quad \left. + |Y_{23}| |V_3| \sin(\delta_3^0 + \gamma_{23} - \delta_2^0) + |Y_{25}| |V_5| \sin(\delta_5^0 + \gamma_{25} - \delta_2^0) \right] \\ &= -2448 \text{ p.u.} \end{aligned}$$

The value

$$\begin{aligned} V_2' &= \frac{1}{Y_{22}} \left[\frac{P_2 - jQ_2'}{(V_2^0)^*} - Y_{21} V_1 - Y_{23} V_3 - Y_{25} V_5 \right] \\ &= 1.0555 \angle 5.11 \end{aligned}$$

But $|V_2'| = |V_2|_{spec}$ which is 1.02 $\angle 5.11$

$$V_3' = \frac{1}{Y_{33}} \left[\frac{P_3 - jQ_3}{(V_3^0)^*} - Y_{32} V_2 - Y_{34} V_4 \right] = 0.98 \angle 7.6$$

$$V_4' = \frac{1}{Y_{44}} \left[\frac{P_4 - jQ_4}{(V_4^0)^*} - Y_{43} V_3 - Y_{45} V_5 \right] = 0.963 \angle -1.53$$

$$\begin{aligned} V_5' &= \frac{1}{Y_{55}} \left[\frac{P_5 - jQ_5}{(V_5^0)^*} - Y_{51} V_1 - Y_{52} V_2 - Y_{54} V_4 \right] \\ &= 0.9836 \angle -0.04^\circ \end{aligned}$$

If $Q_{2\max}$ is 3 p.u find the value of

$$Q_2, V_2, V_3, V_4, V_5$$

The value of Q_2^1 is .2448. Since ~~min~~
minimum permissible value is .3 we fix

$$Q_2 = \underline{.3 \text{ p.u}}$$

Now consider a set of two eqns

$$f_1(x_1, x_2) = 0$$

$$f_2(x_1, x_2) = 0$$

let x_1^0 and x_2^0 are the initial estimates
 Δx_1^0 and Δx_2^0 are the corrections.

Then

$$f_1(x_1^0 + \Delta x_1^0, x_2^0 + \Delta x_2^0) = 0$$

$$f_2(x_1^0 + \Delta x_1^0, x_2^0 + \Delta x_2^0) = 0$$

Expanding into Taylor's series.

$$f_1(x_1^0, x_2^0) + \Delta x_1^0 \left(\frac{\partial f_1}{\partial x_1} \right)^0 + \Delta x_2^0 \left(\frac{\partial f_1}{\partial x_2} \right)^0 + \dots = 0$$

$$f_2(x_1^0, x_2^0) + \Delta x_1^0 \left(\frac{\partial f_2}{\partial x_1} \right)^0 + \Delta x_2^0 \left(\frac{\partial f_2}{\partial x_2} \right)^0 + \dots = 0$$

Now Truncating the series after the first two terms.

$$\begin{bmatrix} f_1(x_1^0, x_2^0) \\ f_2(x_1^0, x_2^0) \end{bmatrix} + \underbrace{\begin{bmatrix} \left(\frac{\partial f_1}{\partial x_1} \right)^0 & \left(\frac{\partial f_1}{\partial x_2} \right)^0 \\ \left(\frac{\partial f_2}{\partial x_1} \right)^0 & \left(\frac{\partial f_2}{\partial x_2} \right)^0 \end{bmatrix}}_{\text{Jacobian}} \begin{bmatrix} \Delta x_1^0 \\ \Delta x_2^0 \end{bmatrix} \approx 0$$

$$\begin{bmatrix} f_1(x_1^0, x_2^0) \\ f_2(x_1^0, x_2^0) \end{bmatrix} = -[J] \begin{bmatrix} \Delta x_1^0 \\ \Delta x_2^0 \end{bmatrix}$$

$$\text{or } \begin{bmatrix} \Delta x_1^0 \\ \Delta x_2^0 \end{bmatrix} = [J]^{-1} \begin{bmatrix} f_1(x_1^0, x_2^0) \\ f_2(x_1^0, x_2^0) \end{bmatrix}$$

$$x_1^1 = x_1^0 + \Delta x_1^0$$

$$x_2^1 = x_2^0 + \Delta x_2^0$$

$(k+1)^{th}$ estimates of roots are

$$x_1^{k+1} = x_1^k + \Delta x_1^k$$

$$x_2^{k+1} = x_2^k + \Delta x_2^k$$

$$\text{where } \begin{bmatrix} \Delta x_1^k \\ \Delta x_2^k \end{bmatrix} = -[J]^{-1} \begin{bmatrix} f_1(x_1^k, x_2^k) \\ f_2(x_1^k, x_2^k) \end{bmatrix}$$

Now consider a system of n eqns.

$$f_1(x_1, x_2, \dots, x_n) = 0$$

$$f_2(x_1, x_2, \dots, x_n) = 0$$

$$\vdots$$

$$f_n(x_1, x_2, \dots, x_n) = 0$$

Let $x_1^0, x_2^0, \dots, x_n^0$ be the initial estimates of unknowns

$\Delta x_1^0, \Delta x_2^0, \dots, \Delta x_n^0$ are the corrections

$$\begin{bmatrix} f_1(x_1^0, x_2^0, \dots, x_n^0) \\ f_2(x_1^0, x_2^0, \dots, x_n^0) \\ \vdots \\ f_n(x_1^0, x_2^0, \dots, x_n^0) \end{bmatrix} + \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} & \dots & \frac{\partial f_1}{\partial x_n} \\ \frac{\partial f_2}{\partial x_1} & \frac{\partial f_2}{\partial x_2} & \dots & \frac{\partial f_2}{\partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial f_n}{\partial x_1} & \frac{\partial f_n}{\partial x_2} & \dots & \frac{\partial f_n}{\partial x_n} \end{bmatrix} \begin{bmatrix} \Delta x_1^0 \\ \Delta x_2^0 \\ \vdots \\ \Delta x_n^0 \end{bmatrix} \approx \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

$$\text{or } f^0 + J^0 \Delta x^0 \approx 0$$

$$\text{or } \Delta x^0 = -[J^0]^{-1} f^0$$

The first estimate of roots is

$$x_1^1 = x_1^0 + \Delta x_1^0$$

$(k+1)^{th}$ root $\Rightarrow$

$$x^{k+1} = x^k + \Delta x^k$$

continue till

$$x^{k+1} - x^k \leq \epsilon$$

Find the value of x in eqn

$$2x - \log x = 7$$

Sol

$$f(x) = 2x - 0.43429 \ln x - 7 = 0$$

$$\frac{df(x)}{dx} = 2 - \frac{0.43429}{x}$$

$$\text{Let } x^0 = 1$$

$$f(x)^0 = -5$$

$$\left(\frac{df(x)}{dx} \right)^0 = 2 - 0.43429 = 1.56571$$

$$\cancel{x^1} = \cancel{x} - \frac{df(x)}{dx}$$

$$\Delta x^0 = - \left[\frac{f(x)^0}{\frac{df(x)}{dx}} \right]$$

$$x^1 = x^0 + \Delta x$$

$$= 1 - \frac{-5}{1.56571} = 4.1934394$$

$$f(x)^1 = 2(4.1934394) - 0.43429 \ln 4.1934394 - 7 = 0.7643149$$

$$\left(\frac{df(x)}{dx} \right)^1 = 2 - \frac{0.43429}{4.1934394} = 1.896435847$$

$$x_2 = 4.1934394 - \frac{0.7643149}{1.896435847} = 3.7904124$$

$$f(x)^2 = 2(3.7904124) - 0.43429 \ln(3.7904124) - 7 = 0.002144308$$

$$\left(\frac{df(x)}{dx} \right)^2 = 2 - \frac{0.43429}{3.7904124} = 1.885424$$

$$x_3 = 3.7904124 - \frac{0.002144308}{1.885424} = 3.789274$$